



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science
Faculty of Science and Technology**

CSC 2210: OBJECT ORIENTED PROGRAMMING 2

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Section: L

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Project Report On *IT Firm Management System*

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total =
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

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1. INTRODUCTION

The IT Firm Management System is a comprehensive, role-based application designed to streamline the operations of an IT organization by integrating project management, client handling, employee administration, and financial tracking into a single unified platform. At its foundation, the system ensures secure and personalized access for every user—whether Admin, Employee, or Client—through a unique UserID that manages login credentials and directs them to their respective dashboards. This role-based structure not only enhances security but also guarantees that each user interacts only with the features relevant to their responsibilities, thereby reducing complexity and minimizing errors. Admins serve as the central authority, managing critical functions such as project approvals, employee assignments, invoice generation, salary records, and asset allocation. Employees, on the other hand, are provided with limited but task-oriented access, enabling them to view assigned projects, update progress, check salary details, and report issues through tickets. Clients interact with the system by submitting project requests, viewing invoices, making payments, and tracking project status. By linking these roles through structured workflows, the system ensures that project requests flow seamlessly from clients to admins, then to employees, with financial and operational records automatically maintained at each stage. Beyond project workflows, the system also incorporates modules for issue tracking, resource management, and salary calculation. Tickets allow employees and clients to report problems, which are then reviewed and resolved by admins. Assets are tracked and allocated to employees as needed, ensuring proper resource utilization. Salaries are managed transparently, with base pay and bonuses recorded to calculate total earnings. Together, these features create a robust management solution that improves efficiency, accountability, and transparency across the organization.

2. FEATURE LIST

2.1 Admin

User Management

- **Add:** Create new users, employees, and clients via the signup form.
- **Delete:** Remove users, employees, and clients completely from the system.

Employee Management

- **Update:** Modify employee **joining date** only by viewing their profile.
- **Delete:** Remove the entire employee record; associated UserID is also deleted to prevent login.

Client Management

- **Delete:** Remove the entire client record; associated UserID is also deleted to prevent login.

Asset Management

- **Update:** Edit asset name, status, and assigned EmployeeID.
- **Delete:** Remove the entire asset record.

Finance & Invoices

- **Add:** Create new invoices with ProjectID, ClientID, Amount, Payment Status, Issue Date, and Due Date.
- **Update:** Modify Amount and Payment Status of existing invoices.
- **Delete:** Remove the entire invoice record.

Issue Tracking

- **Update:** Change ticket status.
- **Delete:** Remove tickets only if their status is marked as Solved.

Salary Management

- **Add:** Assign Base Salary and Bonus for employees.
- **Update:** Modify existing Salary and Bonus values.
- **Delete:** Remove the entire salary record for a selected employee.

2.2 Employee

Profile

- **View:** Displays First Name, Last Name, Designation, and Joining Date.
- **Update:** Employees can update First Name, Last Name, and Designation.
- **Joining Date:** Assigned by Admin after profile update.

Projects

- **View:** Shows assigned project details (Project Name, Status, ClientID).
- **Update:** Employees can update project status based on progress; otherwise remains “Pending.”

Assets

- **View:** Employees can view allocated assets and their status (managed by Admin).

Salary

- **View:** Employees can view Base Salary and Bonus.
- **Total Salary:** System calculates and displays total (Base + Bonus).

Tickets

- **Add:** Employees can create tickets with issue description.
- **Default Status:** Tickets automatically marked “Pending” until Admin updates.

2.3 Client

Profile

- **View:** Displays company name, contact number blank and email shown when signed up.
- **Add:** Clients will add their contact details.

Project Requests

- **Add:** Clients can request new projects by giving projectName.
- **View:** Clients can view approved projects and assigned employees which are managed by admin.

Invoices

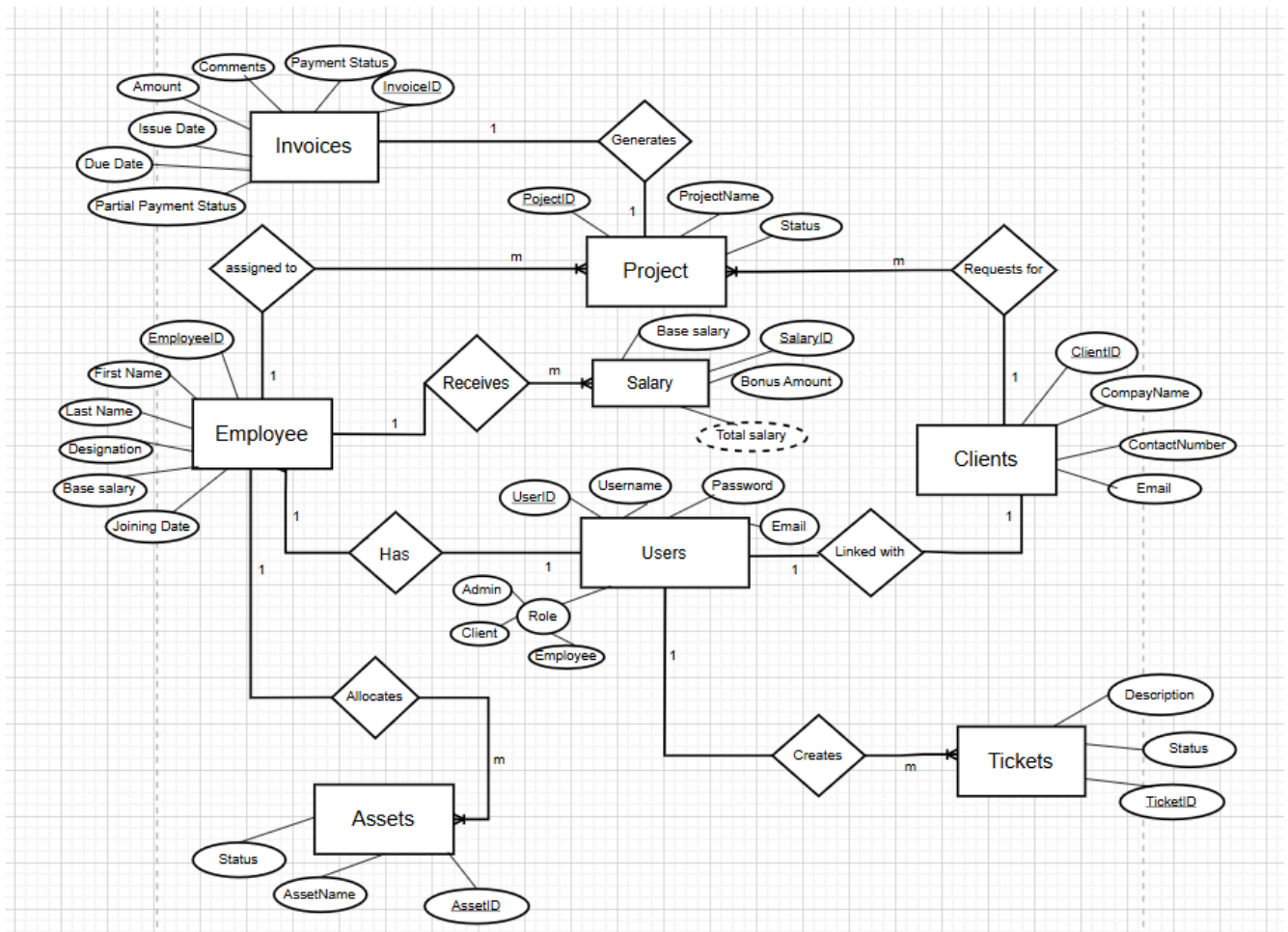
- **View:** Clients can view invoices related to their projects (Amount, Payment Status, Due Date).

- **Add:** Clients can add Partial Payment Amount and add Comments if necessary.

Tickets

- **Add:** Clients can raise tickets for issues or requests.
- **View:** Clients can track the status of tickets, which initially shows Pending and is later updated by the Admin.

3.ER DIAGRAM



The IT Firm Management System is built on a role-based foundation, ensuring secure and personalized access for Admins, Employees, and Clients. Each user is identified by a unique UserID, which manages login credentials and directs them to the correct dashboard. Employees are tracked by an EmployeeID, storing their designation, joining date, and salary details. Clients are managed through a ClientID, maintaining company information and contact details. At the core lies a streamlined project workflow: Clients submit project requests, Admins approve and assign them to employees, and the system generates a unique InvoiceID for billing. Invoices record payment status, issue date, and due date, ensuring financial accuracy. Beyond projects, the system supports issue tracking through tickets, managed by Admins, and asset allocation, where company resources are assigned to employees. Finally, the Salary entity ensures transparent pay by recording base salary and bonuses, automatically calculating total earnings.

3.1 Normalization:

Users->has->employee:

UNF: UserID, Username, Password, Role, EmployeeID, FirstName, LastName, Designation, BaseSalary, JoiningDate

1NF: UserID, Username, Password, Role, EmployeeID, FirstName, LastName, Designation, BaseSalary, JoiningDate

2NF:

1. Users: UserID(PK), Username, Password, Role
2. Employees: EmployeeID (pk), FirstName, LastName, Designation, BaseSalary, JoiningDate, UserID(FK))

Users->linked with ->client:

UNF: UserID, Username, Password, ClientID, CompanyName, ContactNumber, Email

1NF: same as UNF

2NF:

1. Users: UserID PK, Username, Password
2. Clients: ClientID PK, CompanyName, ContactNumber, Email, UserID FK

Client->requests for->project:

UNF: ClientID, CompanyName, ProjectID, ProjectName, Status

1NF: same as UNF

2NF:

1. Clients: ClientID PK, CompanyName
2. Project: ProjectID PK, ProjectName, Status, ClientID FK

Employee->assigned to-> project:

UNF: EmployeeID, EmployeeName, ProjectID, ProjectName, Status

1NF: same as UNF

2NF:

1. Employee: EmployeeID PK, FirstName, LastName
2. Project: ProjectID PK, ProjectName, Status, EmployeeID FK

Employee->receives->salary:

UNF: EmployeeID, EmployeeName, SalaryID, BaseSalary, BonusAmount, TotalSalary

1NF: Derived: TotalSalary

2NF:

1. Employee: EmployeeID PK, FirstName, LastName
2. Salary: SalaryID PK, BaseSalary, BonusAmount, EmployeeID FK

Project-> generates -> invoices:

UNF: ProjectID, ProjectName, InvoiceID, Amount, PaymentStatus, IssueDate, DueDate, PartialPaymentAmount, Comments

1NF: same as UNF

2NF:

1. Project: ProjectID PK, ProjectName
2. Invoices: InvoiceID PK, Amount, PaymentStatus, IssueDate, DueDate, PartialPaymentAmount, Comments, ProjectID FK

Employee->allocates->assets:

UNF: EmployeeID, EmployeeName, AssetID, AssetName, Status

1NF: same as UNF

2NF:

1. Employee: EmployeeID PK, FirstName
2. Assets: AssetID PK, AssetName, Status, EmployeeID FK

Users->creates->tickets:

UNF: UserID, Username, TicketID, Description, Status

1NF: same as UNF

2NF:

1. **Users:** UserID PK, Username
2. **Tickets:** TicketID PK, Description, Status, UserID FK

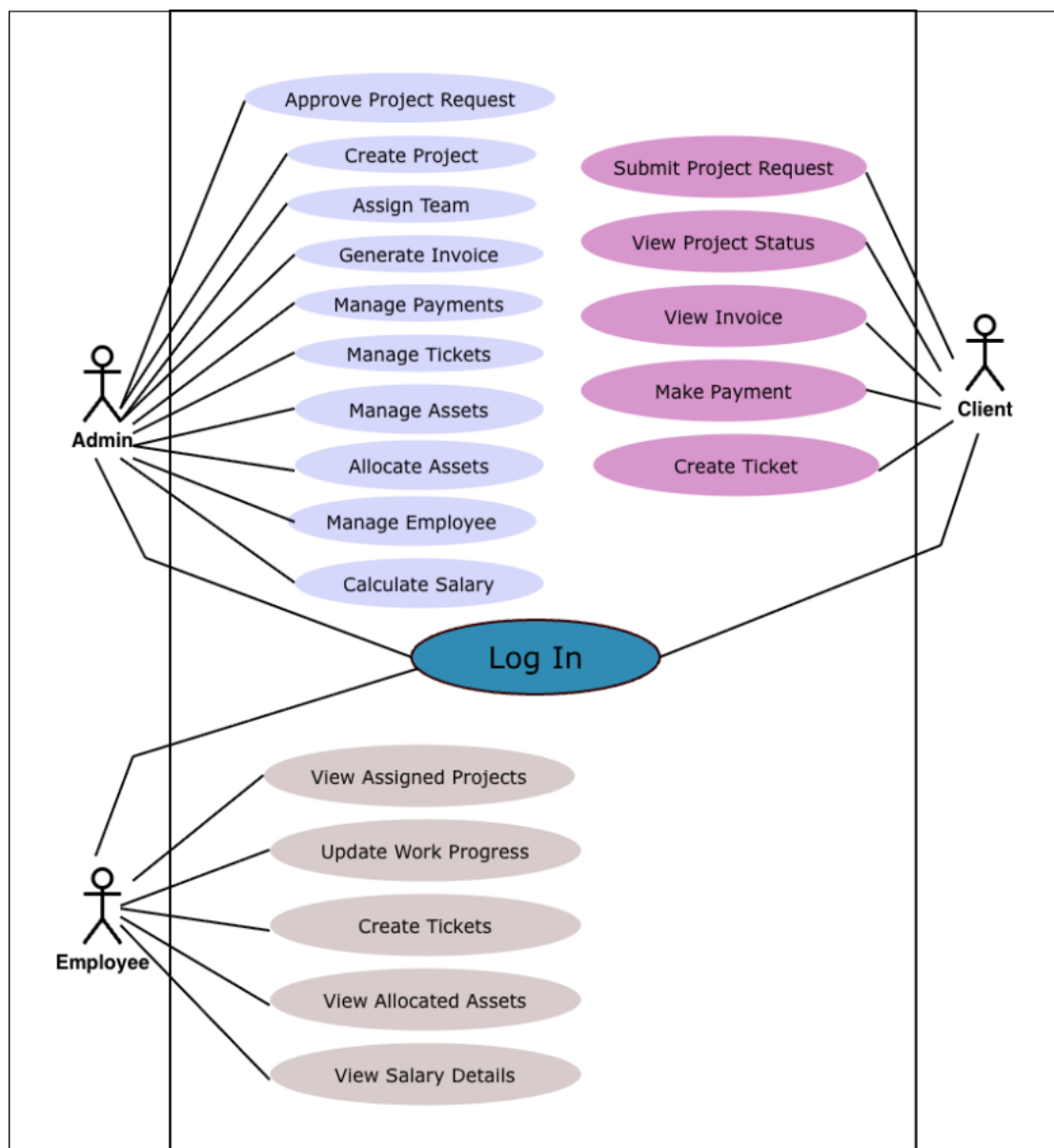
3.2 Finalization:

- 1) **Users:** UserID PK, Username, Password, Role
- 2) **Employee:** EmployeeID PK, FirstName, LastName, Designation, JoiningDate, UserID FK
- 3) **Clients:** ClientID PK, CompanyName, ContactNumber, Email, UserID FK
- 4) **Project:** ProjectID PK, ProjectName, Status, EmployeeID FK, ClientID FK
- 5) **Salary:** SalaryID PK, BaseSalary, BonusAmount, EmployeeID FK
- 6) **Invoices:** InvoiceID PK, Amount, PaymentStatus, IssueDate, DueDate, PartialPaymentAmount, Comments, ProjectID FK
- 7) **Assets:** AssetID PK, AssetName, Status, EmployeeID FK
- 8) **Tickets:** TicketID PK, Description, Status, UserID FK

4. UML DIAGRAM

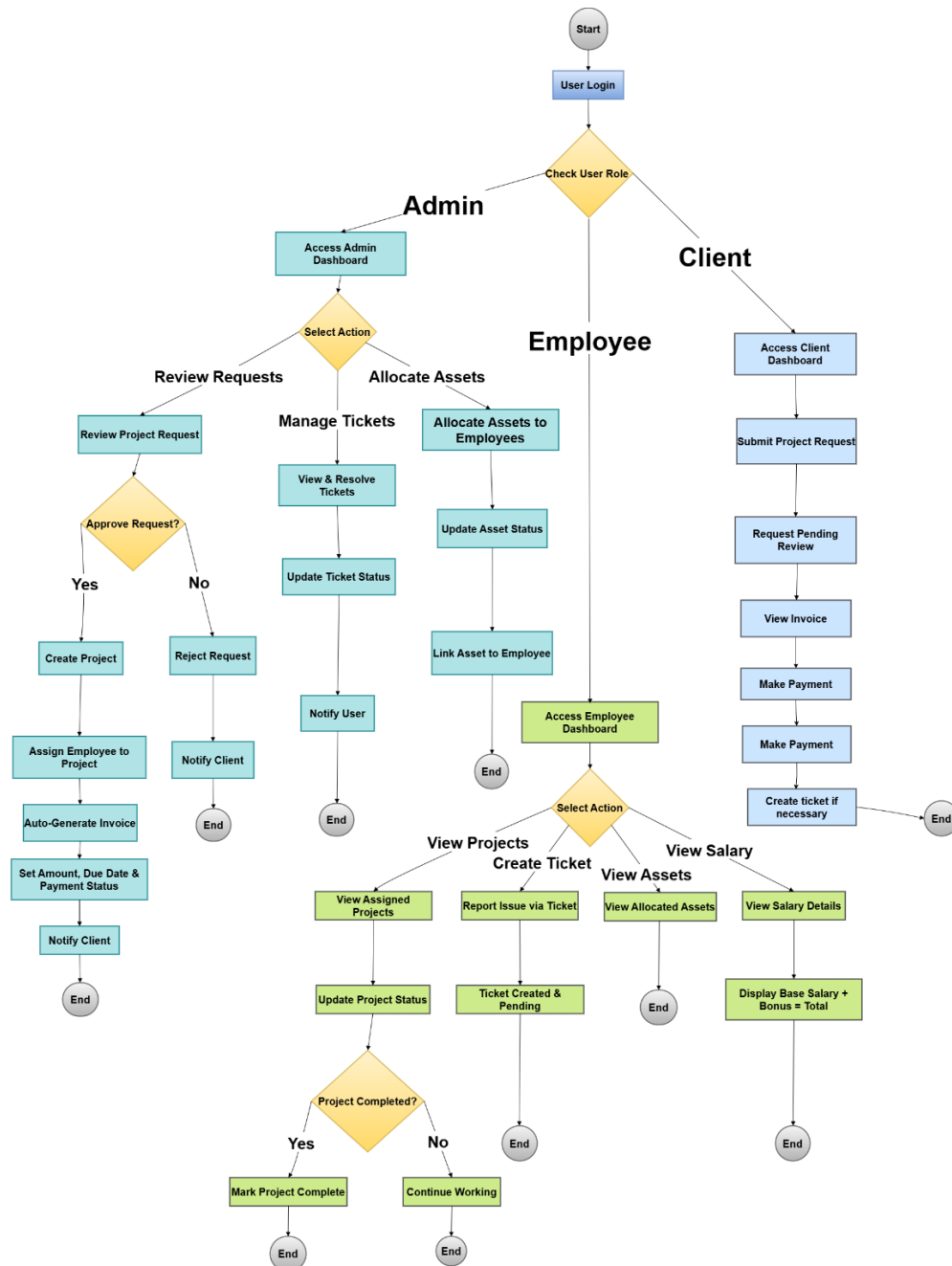
4.1 Use Case Diagram

This use case diagram shows how Admin, Client, and Employee interact with the system after logging in. The Admin controls the main operations such as approving project requests, creating projects, assigning teams, generating invoices, managing payments, tickets, assets, employees, and salaries. The Client can submit project requests, view project status and invoices, make payments, and create tickets. The Employee can view assigned projects, update work progress, create tickets, view allocated assets, and check salary details. Overall, it represents a role-based system where each user has different responsibilities and permissions.



4.2 Activity Diagram

The activity diagram illustrates the complete workflow of the system starting from user login. Based on the user role (Admin, Client, or Employee), the system provides different activities. The Admin reviews project requests, manages tickets, allocates assets, assigns employees, and controls invoices and approvals. The Client submits project requests and waits for approval. The Employee views assigned projects, updates work progress, manages tickets, and accesses salary and asset information. The diagram shows decision points such as request approval and project completion, ensuring a smooth and role-based operational flow.



5. CONCLUSION

In conclusion, the IT Firm Management System provides a holistic solution for managing the diverse operations of an IT organization. Its role-based foundation ensures that each user—Admin, Employee, or Client—has secure access to the features necessary for their role, while sensitive administrative controls remain restricted to authorized personnel. By integrating project workflows, financial management, ticket handling, asset allocation, and salary processing, the system eliminates redundancy and creates a streamlined environment where information flows smoothly between stakeholders. The system's design emphasizes both functionality and transparency. Clients benefit from clear project tracking and invoice management, employees gain structured access to their tasks and compensation details, and admins maintain full oversight of organizational processes. This balance of control and accessibility strengthens collaboration, reduces operational bottlenecks, and enhances trust among all parties involved. Ultimately, the IT Firm Management System stands as a scalable and reliable platform that not only simplifies day-to-day operations but also supports long-term organizational growth. By combining secure role-based access with integrated modules for projects, finances, assets, tickets, and salaries, it delivers a professional, efficient, and transparent management framework that meets the evolving needs of IT firms.